The empirical recurrence for OEIS A267663: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

2 September 2026

Abstract

OEIS A267663 records "Empirical recurrence of order 80", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 699$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 80, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 80 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A267663 is "Number of n X 4 0..1 arrays with every repeated value in every row unequal to the previous repeated value, and in every column equal to the previous repeated value, and new values introduced in row-major sequential order.". It has offset 1 and begins

$$5, 50, 500, 3143, 18432, 98438, 508681, 2560344, \ \ldots$$

The entry states:

> Empirical recurrence of order 80 (see link above).

As of the "Last modified" line on the live entry (May 04 2021 16:01:48, revision 6) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 699$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' =$ 699.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 1398$ exact terms of the model returns order 80 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{80} c_i\, a(n-i),$$

c_{1}	19	c_{21}	−13178324756523	c_{41}	−7606755311614167	c_{61}	575723327959
c_{2}	−52	c_{22}	−3439747891918	c_{42}	7975238064341320	c_{62}	369080578778
c_{3}	−1158	c_{23}	56651654959073	c_{43}	5374310973325341	c_{63}	−116843549232
c_{4}	7561	c_{24}	757673931636	c_{44}	−6412726776500595	c_{64}	−46918526430
c_{5}	26038	c_{25}	−200476800635794	c_{45}	−3126965731710710	c_{65}	18572142417
c_{6}	−320506	c_{26}	35258432196927	c_{46}	4353872003693079	c_{66}	4572671298
c_{7}	−102454	c_{27}	588326908873800	c_{47}	1469787884889545	c_{67}	−2311928550
c_{8}	7726480	c_{28}	−191765610930443	c_{48}	−2497919012582633	c_{68}	−318279496
c_{9}	−8451182	c_{29}	−1439268350749609	c_{49}	−535540753942576	c_{69}	222795197
c_{10}	−124627505	c_{30}	647037000140120	c_{50}	1210579562754821	c_{70}	12904135
c_{11}	251826526	c_{31}	2945827430945074	c_{51}	134812585302272	c_{71}	−16246414
c_{12}	1446045466	c_{32}	−1637009637603024	c_{52}	−494817219979474	c_{72}	45138
c_{13}	−4050808634	c_{33}	−5056395993626730	c_{53}	−116155353 83311	c_{73}	863263
c_{14}	−12528499455	c_{34}	3293185530674163	c_{54}	170085026093232	c_{74}	−43300
c_{15}	45199727620	c_{35}	7287672581678803	c_{55}	−9052472625893	c_{75}	−31382
c_{16}	82417718627	c_{36}	−5408709842731554	c_{56}	−48940774422125	c_{76}	2756
c_{17}	−379574310510	c_{37}	−8820980396279501	c_{57}	5975464908860	c_{77}	694
c_{18}	−410565527110	c_{38}	7356780558437497	c_{58}	11709571141358	c_{78}	−82
c_{19}	2497153429113	c_{39}	8956411878169088	c_{59}	−2182010436365	c_{79}	−7
c_{20}	1494360135463	c_{40}	−8358349123829429	c_{60}	−2307118944724	c_{80}	1

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 80, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $80+4$ terms removed returns order 80 as well, so $\deg q_{\min} = 80 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 20 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $80+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 80 that this sequence satisfies — and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A267663.

[2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.

[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.